Enzymes, Moonlighting Enzymes, Pseudoenzymes: Similar in Sequence, Different in Function

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Abstract

The function of a newly sequenced protein is often estimated by sequence alignment with the sequences of proteins with known functions. However, members of a protein superfamily can share significant amino acid sequence identity but vary in the reaction catalyzed and/or the substrate used. In addition, a protein superfamily can include moonlighting proteins, which have two or more functions, and pseudoenzymes, which have a three-dimensional fold that resembles a conventional catalytically active enzyme, but do not have catalytic activity. Pseudoenzymes and moonlighting proteins are widespread in the evolutionary tree and are found in many protein families, and they are often very similar in sequence and structure to their monofunctional and catalytically active counterparts. A greater understanding is needed to clarify when similarities and differences in amino acid sequences and structures correspond to similarities and differences in biochemical functions and cellular roles. This information can help improve programs that identify protein functions from sequence or structure and assist in more accurate annotation of sequence and structural databases, as well as in our understanding of the broad diversity of protein functions.
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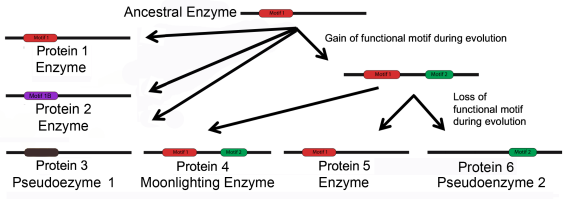

Figure 1 An enzyme superfamily can contain enzymes with canonical catalytic activity, enzymes with noncanonical catalytic activity, pseudoenzymes, and moonlighting enzymes. Some pseudoenzymes evolved from moonlighting proteins. During evolution an enzyme with a catalytic functional motif (motif 1) can gain a second functional motif (motif 2) to become a moonlighting protein. Gene duplication and subsequent evolution can result in subfunctionalization with duplicated genes each losing one functional motif. The resulting homologous proteins in the protein family, which can be in the same or different species, can include proteins that evolved from proteins that never contained the motif 2 (protein 1), a moonlighting protein that has two functions symbolized by motif 1 and motif 2 (protein 4), proteins that evolved from proteins that gained but then lost motif 2 (protein 5), and psudoenzymes that evolved from proteins that gained motif 2 but lost motif 1 (protein 6).

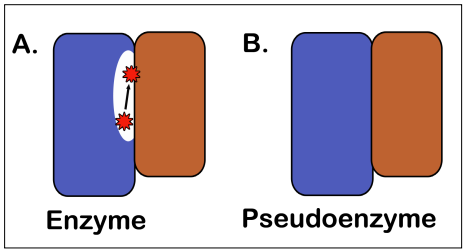

Figure 2. Pseudoenzymes resemble enzymes but have changes in or near the active site that prevent catalytic activity. In traditional pseudoenzymes, the protein function is derived from part of the main function of the ancestral protein. A. In a kinase, transfer of a phosphoryl group (red star) and binding to another protein (brown oval) are aspects of the catalytic activity. B. In a pseudoenzyme related to the kinase in A, the transferase function has been lost, and the protein binding activity is repurposed as the protein activity.

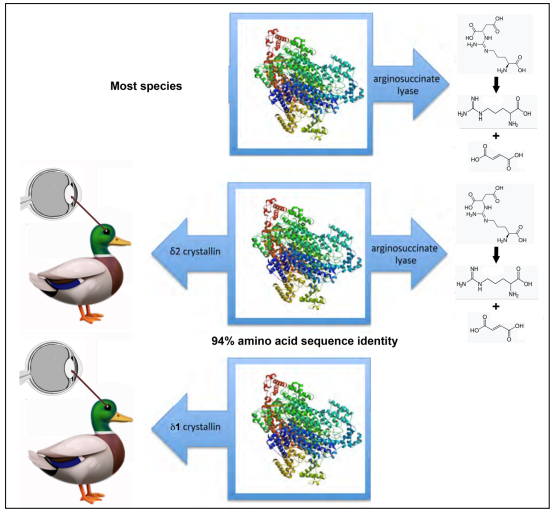

Figure 3. The function of some pseudoenzymes do not appear to be derived from an aspect of the catalytic function.
Top. Arginosuccinate lyase is an active enzyme in many species.
Middle. In ducks, the delta2 crystallin is a crystallin in the lens of the eye and an active enzyme in other tissues. Bottom. Also in ducks, the delta1 crystallin shares 94% amino acid sequence identity with the delta2 crystallin but does not have arginosuccinate lyase catalytic activity.

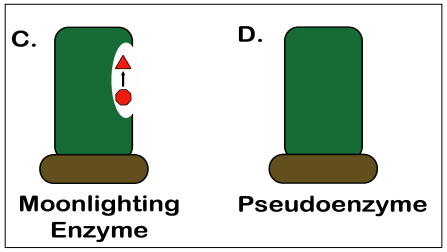

Figure 4. Pseudoenzymes evolved from moonlighting proteins. Some pseudoenzymes arise from moonlighting enzymes that have lost the catalytic function. C. In a moonlighting enzyme, the catalytic function (hexagon converted to triangle) in the active site is separate from and independent of another function, such as binding to another protein (brown oval). D. In a pseudoenzyme evolved from the moonlighting protein, the catalytic activity has been lost, and the remaining activity is different from other members of the enzyme protein family.

Table 1. Pseudoenzymes with Moonlighting Protein homologues

Pseudoenzyme	Moonlighting protein
Duck delta1 crystallin	duck delta 2 crystallin/arginosuccinate lyase
S. cerevisiae Alt2	KlAtl1, LkAtl1 alanine aminotransferases*
S. cerevisiae Gal3	Gal1 galactokinase/transcription regulator
mammalian IRB2	mammalian IRB1/aconitase

* The noncatalytic function(s) of KlAtl1and LkAtl1 might be different from the function of ScAlt2

Summary

- One protein family can contain enzymes, moonlighting enzymes, and pseudoenzymes.
- Many pseudoenzymes have functions that share some aspect of the function of catalytically active homologues
- But – some pseudoenzymes have a very different function from their catalytically active homologues
- These latter proteins might have evolved from ancestral moonlighting proteins.
- Comparison of predicted pseudoenzymes (that resemble enzymes but lack the conserved catalytic amino acids) with known moonlighting proteins might help in the development of testable hypotheses of functions for some of these potential pseudoenzymes.